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Monsoon, 2025

B. Tech 5th Semester Examination

Hydrology and Water Resources Engineering

Course Code: BCE23504T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

1. Choose the correct answer.

(1 x 10 = 10)

(i) The process by which the water enters the ground is called-

- (a) Percolation
- (b) Infiltration
- (c) Both (a) and (b)
- (d) None of the above

(ii) In the history of Hydrology, the period of Speculation was from-

- (a) Prior to AD 1400
- (b) 1400 to 1600
- (c) 1600 to 1700
- (d) 1950 to till date

(iii) The double mass curve technique is adopted to-

- (a) Check the consistency of raingauge records
- (b) Estimate the missing rainfall data
- (c) Find the average rainfall over a large number of years
- (d) Find the cumulative rainfall occurring in different months

(iv) A very thin fog is termed as-

- (a) Snow
- (b) Mist
- (c) Hail
- (d) Frost

(v) A hydrograph is a plot of-

- (a) Rainfall intensity against time
- (b) Stream discharge against time
- (c) Cumulative rainfall against time
- (d) Cumulative runoff against time

(vi) The term base flow denotes-

- (a) Delayed groundwater flow reaching a stream
- (b) Delayed groundwater and snowmelt reaching a stream
- (c) Delayed groundwater and interflow
- (d) The annual minimum flow in a stream

(vii) A geological formation which is essentially impermeable for flow of water even though it may contain water in its pores is called-

- (a) Aquifer
- (b) Aquifuge
- (c) Aquitard
- (d) Aquiclude

(viii) The discharge per unit drawdown at a well is known as-

- (a) Specific yield
- (b) Specific storage
- (c) Safe yield
- (d) Specific capacity

(ix) A hydropower project has-

- (a) High tax
- (b) Less tax
- (c) No tax
- (d) None

(x) Total project cost consists of-

- (a) Separable Cost
- (b) Joint Cost
- (c) Both Separable and Joint Cost
- (d) Remaining Benefit Methods

2. Explain the schematic representation of the hydrologic cycle.

OR

(6)

Explain the forms of precipitation.

3. For a drainage basin of 600 km^2 , isohyets drawn for a storm gave the following data:

Isohyets (interval) (cm)	15-12	12-9	9-6	6-3	3-1
Inter-isohyetal area (km^2)	92	128	120	175	85

Estimate the average depth of precipitation over the catchment.

OR

(5)

Using Thiessen Polygon Method, compute the depth of average precipitation over a basin using the following data:

Raingauge station	A	B	C	D	E	F	G	H
Thiessen Polygon area (km^2)	720	1380	1440	1040	900	2220	419	1456
Recorded rainfall (mm)	135	143	137	128	102	115	99	101

4. A 7 hour storm over the basin of 1830 km^2 produced the rainfall intensity at half an hour interval are 4, 9, 20, 18, 13, 11, 12, 2, 8, 16, 17, 13, 6 and 1 mm/hour. If the corresponding observation run-off is $73.2 \times 10^6 \text{ m}^3$, estimate the ϕ - index of the storm.

OR

(5)

Using the knowledge of hydrology, discuss how infiltration is measured.

5. The 3-hour unit hydrograph for a basin has the following ordinates. Determine the 9-hour unit hydrograph ordinates of the basin by the **Method of Superposition**.

Time (hr)	0	3	6	9	12	15	18	21	24	27	30	33
3-hr UH Ordinates (m^3/sec)	0	15	60	100	130	100	75	42	23	12	5	0

OR

(8)

Analyze the above question using **S- Curve Method**.

6. A basin has an area of 400 km^2 and the following characteristics:

Basin Length = 35 km

Length upto the centroid of the basin = 10 km

Snyder's Coefficient: $C_t = 1.5$, $C_p = 0.70$

Analyze synthetically the 3-hour synthetic unit hydrograph for this basin using Snyder's method.

OR

(7)

Show confined and unconfined aquifers with the help of a diagram. A 30 cm diameter well completely penetrates a confined aquifer of permeability 45 m/day. The length of the strainer is 20m. Under steady state of pumping, the drawdown at the well was found to be 3.0m and the radius of influence was 300m. Estimate the discharge.

7. Differentiate between Hydropower and Thermalpower.

OR

(4)

Summarize the social responsibilities of water resource engineering.

8. Using the data given below, solve the allocation to each project purpose by the **Remaining Benefit Method**.

Total project cost = Rs. 1765 Cr.

Project function	Separable cost (Crores Rs.)	Estimated Benefits (Crores Rs.)	Alternate single purpose cost (Crores Rs.)
Flood mitigation	360	450	500
Hydropower	620	1550	900
Irrigation	100	300	620
Navigation	100	150	60

OR

(5)

Solve the above question using **Alternative Justifiable Expenditure Method**.